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Adaptive Modelling of EEG Signals to Produce Accurate Time-Frequency Decompositions for Use in BCI

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Abstract*— Motor imagery and actual movement are both tasks that bring forth a noticeable change in the subject's EEG mu rhythm known as Even-Related Desynchronisation (ERD). They appear as magnitude decreases of the frequencies included in the said band and can be tracked and measured for the automatic real-time detection and classification of the events. It has been proven that an important percent of the changes happen within a narrow frequency band called a reactive band, providing thus the means to significantly improve the efficiency of the interpretation of such events by concentrating the decisive information. The algorithm presented in this paper automatically identifies a subject's specific reactive band by detecting the highest decrease in power. The decision is made after the recorded EEG signal is modeled with a Band-Limited Multiple Fourier Linear Combiner (BMFLC). The method adaptively estimates the amplitude of each frequency component in the given band of interest and produces a precise time-frequency map that can be afterwards used for increasingly accurate classification and BCI applications.

***Keywords*— motor imagery; mu-rhytm; ERD detection; adaptive estimation; BCI; BMFLC; time-frequency map**

I. INTRODUCTION

Motor tasks such as actual movement or motor imagery, are stimulus that induce a form of time-locked activity visible also in the waveform of the EEG waves but very easily noticeable in the frequency representation. Motor tasks cause strong ERD (Event Related Desynchronisations) visible in the frequency band spanning between 6 and 14 Hz called the μ band; but each person presents a specific subband that concentrates most of their sensorimotor ERD.

These desynchronisation have received considerable attention due to their use in BCI (Brain Computer Interface). These systems find usage in an increasing number of domains, of which prosthetics control and rehabilitation have proved of great interest. The most used methods deployed in the investigation of such events are STFT (Short Time Fourier Transform), the wavelet transform and the BMFLC (Band-limited Fourier Linear Combiner) which builds a model for the signal using Fourier series. These methods each present certain disadvantages, the STFT has a weak frequency resolution, the wavelet transform partly solves this issue but is computationally very costly [1] and the BMFLC algorithm has poor performance on its own and is very dependent on the parameters governing the LMS method used. Perhaps the strongest point of the BMFLC method is its flexibility, and while in basic form this algorithm has unsatisfactory performances, a great number of modifications can be made to greatly improve the obtained performance. This paper studies the impact of the brought variations upon the quality of the BMFLC method and compares them with the final purpose of identifying the most efficient solution from the viewpoints of both performance and duration.The model obtained from by the use of BMFLC is then intended for use in BCI applications as it provides a very precise time frequency map to work with.

II. DATASET DESCRIPTION

The datasets used for developing and testing these algorithms were recorded in 2008 by the staff of Graz University of Technology, Austria for the fourth BCI competition. The EEG data was monopolarly measured with electrodes positioned as per the 10-20 international system and considers 9 subjects measurements. The tasks included four different classes of movement conducted accordingly with the provided cues of which only those related to the upper limb movement were selected for this study (left hand, right hand) [2]

The data was pre-processed using the available EEGLab toolbox with MATLAB software. The first step was band-pass filtering the data between 6 and 14 Hz using a 500 order FIR filter as to avoid affecting the phase of data. The high order of the filter is acceptable because the EEG is filtered before segmentation and because the recordings begin with an EOG evaluation period so it does not cause relevant loss while the filter loads. The trials for each hand were then extracted from the data and epoched with the time-locking event of the respective cue in order to assure synchronization between trials. Each

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extracted trial is 6 seconds long covering a 2 seconds period before the cue and a 4 seconds period after.

III. Methods

A. Band Limited Multiple Fourier Linear Combiner

The BMFLC method assumes that the EEG signal can be represented as a sum of simple sine and cosine functions of various frequencies and attempts to find the amount in which each component contributes at a given moment in time [1], [3]. The method uses the LMS adaptive algorithm to minimize the square error, it is a steepest descent gradient search which works iteratively and has the advantages of being simple, computationally efficient and widely applicable. Its issues reside in convergence due to noise and initialization of the values [4]. In this case the LMS works in estimating the weights **w** corresponding to the sine and cosine component of the Fourier series identified for the given signal.

The band of interest $[\omega_1\text{-}\omega_n]$ is divided into a number of n subbands of length $\Delta\omega$ to form the BMFLC, ω_p being the p^{th} frequency calculated as $\omega_p=2\pi f/f_s$ where f and f_s are the respective frequency and the sampling frequency (Hz). Looking at the expression below, for an incoming EEG signal, s_k denotes the value at sampling time k, where $k \epsilon [1, m]$, and y_k denotes the model obtained using this method. For the model, a_{pk} and b_{pk} represent the adaptive weights corresponding to ω_p and sampling moment k.

$$y_{k=} \sum_{p=1}^{n} a_{pk}\sin(\omega_p k) + b_{pk}\cos(\omega_p k) \qquad (1)$$

Let $\mathbf{x}_k$ and $\mathbf{w}_k$, column vectors, be the input reference vector and the adaptive weight vector, at time instance k.

$$\mathbf{x}_k = [\sin(\omega_1 k) \dots \sin(\omega_n k)\ \ \cos(\omega_1 k) \dots \cos(\omega_n k)]^T (2)$$

$$\mathbf{w}_k = [a_{1k} \quad a_{2k} \quad \dots \quad a_{nk}\ ,\ b_{1k} \quad b_{2k} \quad \dots \quad b_{nk}]^T$$

The algorithm respects the following steps:

$$y_k = \mathbf{w}_k^T \mathbf{x}_k \qquad (3)$$

$$\varepsilon_k = s_k - y_k$$

$$\mathbf{w}_{k+1} = \mathbf{w}_k + 2\eta\varepsilon_k^T \mathbf{x}_k$$

We can compute the absolute weight vector $\mathbf{w}_k^f$ which includes all frequency components at time instant k.

$$\mathbf{w}_k^f = \left[\frac{\sqrt{a_{1k}^2+b_{1k}^2}}{2} \quad \dots \quad \frac{\sqrt{a_{nk}^2+b_{nk}^2}}{2}\right]^T \qquad (4)$$

The time frequency decomposition which describes the entire signal is the weight matrix D that can be expressed as:

$$\mathbf{D} = \left[w_1^f \quad \dots \quad w_k^f \quad \dots \quad w_m^f\right]. \qquad (5)$$

This matrix offers easy access to the exact values regarding all frequency components. Since $\mathbf{D}(k,f)$ can be recognized as the amplitude of the frequency component f at time instant k, the signal's power can be calculated as $\mathbf{P}(k,f)= \mathbf{D}(k,f) \times \mathbf{D}(k,f)$. Thus, the power distribution **P** of the signal can be computed for all components by squaring the matrix **D** element by element.

It has been proven by Pfurtscheller et al [5] that ERD is more successfully detected when using the selected reactive band rather than the entire mu-band. In order to identify this reactive band we must observe the highest difference in power between the rest and imagery periods which can be measured by observing the variations of the weights.

B. Normalized least mean square

Normalized least mean square (NLMS) is an improvement brought to the "pure" LMS method which is very sensitive to the scaling of its input. This causes convergence problems due to the magnitude imbalance and unpredictability of the input. A very simple way of overcoming this problem and controlling the adjustment process is by normalizing with the input power. The weight adjustment equation becomes:

$$\mathbf{w}_{k+1} = \mathbf{w}_k + \frac{2\eta\varepsilon_k^T \mathbf{x}_k}{\|x_k\|^2},$$

where $\|x_k\|^2$ represents the input signal energy. This approach is very useful when dealing with the unaltered input, which originally ranges between -10 and 10. When the Laplacian filter is used, the input has a value closed to the learning rate and the normalization proves damaging to the results.

C. Small Laplacian Filter

The EEG signal recorded with each electrode is a summation of the potentials generated by the area directly underneath the electrode and the electrical activity of the neighboring areas. In order to retain only the most meaningful variations and eliminate the continuous component, the mean of the signals collected by the respective neighboring electrodes is subtracted from the EEG channels corresponding to the C3 and C4 sites, the left and right regions of the motor cortex [6]. This small Laplacian filter, as defined by McFarland et al has the following formula

$$V_{CH}^{LAP} = V_{CH}^{ref} - \frac{1}{4}\sum_{j \epsilon S_{CH}} V_j^{ref},$$

where S_{CH} is the set of nearest neighbor electrodes of the point of interest CH.

D. Forcefully Adjusting the Weights

Another step is added to the algorithm that evaluates the error at the previous iteration ε_{k-1}. If the absolute value is above a decided threshold level (th), the weights are decreased with a chosen percent pr.

$$w_k = w_k - pr \times w_k$$

Both values of the threshold and of the percent *pr* are case specific and determined by observing the magnitude levels of the errors. The advantage of this method is that it is very effective and has a low computational cost which makes it ideal for BCI applications.

E. Adding a learning step

To overcome the damage done by the high error values, another approach is presented. Inspired by the functioning of neural networks based on the same algorithm, a learning step has been introduced. In this case, the LMS algorithm is no longer applied to one entire trial but to a selected moving window. Two variations of this idea have been selected for presentation, the first using a one second long moving window and the other using a half-second moving window. At the end of the training period the obtained weights saved in matrix **D** are used to calculate the model and the corresponding errors.

IV. RESULTS

The BMFLC is applied separately on each trial and the resulting weight matrices are averaged to obtain the final matrix **D** for a certain subject. In order to evaluate the accuracy (%) of the model we define:

$$Accuracy = \frac{RMS(s) - RMS(\varepsilon)}{RMS(s)} \times 100,$$

where $RMS(s) = \sqrt{\sum_1^n s/n}$, n being the number of trials per subject. The statistical mean and standard deviation of the accuracy parameter presented in the Table I, are the decisive criteria selecting the most relevant methods.

The two general improvement methods: NLMS and Laplacian filter were meant to lay a better ground for the following two methods. Surprisingly, as it can be observed by studying the results in Table I ,the NLMS becomes redundant and nocive when the Laplacian is applied, decreasing the accuracy of the model drastically. However, when working with the brute signal, the NLMS is useful in reducing the oscillations of the resulting errors over the span of both the entire trial and the window of chosen length.

The Laplacian filter pre-processes the data enhancing the Signal-to-Noise ratio and removing the mean component [6]. This is very inconvenient when dealing with a method as sensitive as the LMS, as a parasite continuous mean, would increase error magnitude and cause the method to fail in producing a reliable result. Visualizing the time-frequency decomposition matrix as an image proves that the quality of time variations noticeably improves, suggesting a more realistic model.

While conducting this study, it has been noticed that at the moment of desynchronisation, the sudden and acute loss of stationarity in the recorded EEG signal is the trigger that induces great errors in the model. These relative errors are too big for the LMS algorithm to compensate causing the model to fail by obliterating the ERD. Thus, the need to quickly manage these unbalancing errors has appeared. While adapting the learning rate has proven insufficiently powerful, two other methods are proposed in this paper. The two strategies differ in the way of applying the LMS algorithm.

One approach treats the trials as a whole and overcomes the errors by forcefully adjusting the weights as presented earlier at section *D*. The method has two remarkable advantages, it is very fast and undemanding as it implies little extra calculations by comparison to the simple LMS. These values assured a constant suppression of the errors and proper coherence for the model. With the increase in threshold value and the decrease in parameter pr the errors accumulate and the quality of the model reduces. Surprisingly, an even lower threshold also introduces errors and so does escalating the value of pr.

The approach described at section *E*. uses a moving window method and applies the LMS algorithm on each fragment repeatedly, minimizing the error with each iteration. At the end of the run, the weights for the entire window are added to **D**. Given that the stationarity period for EEG signals is estimated to one second, but also taking into consideration the studied data which is known to present important variations, the selected window durations are one and 0.5 seconds. The number of repetitions is chosen after running the LMS algorithm on a simulated signal consisting of a small number of known sinusoids and observing the time elapsed until convergence. The minimum number of iterations can be calculated as the time span thus obtained, divided by the duration (in seconds) of the window. The reasoning safely assumes that the time necessary in order to obtain a satisfactory result for a signal as complex as the EEG is at least equal to the value given by the mentioned simulation.

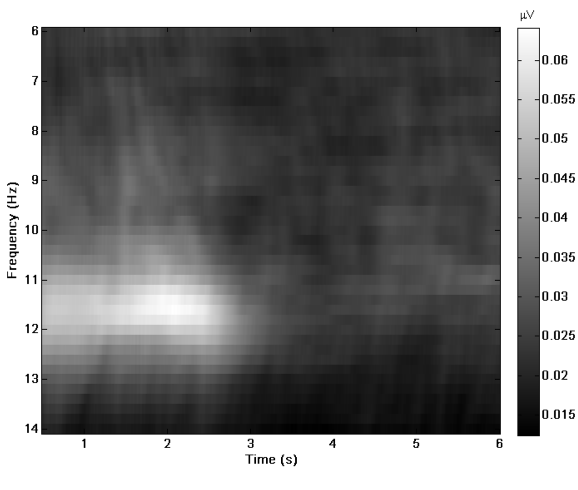

Fig. 1: Weight matrix as colormap averaged for subject 3, left hand motor imagery. The method used is D. which treats the entire trial

After observing the methods studied so far, a hybrid approach is proposed, a combination of the ideas presented in sections *D.* and *E.*, which includes the forced minimization of the errors by weight adjustment in the additional learning step. This approach has proved most effective improving the mean relative error of the data simultaneously with decreasing the standard deviation considerably by comparison with the other implementations. The results are summarized in the adjacent Table I where *D.* refers to the forceful adaptation of the weights, and *E.* includes the introduction of a learning step .

TABLE 1 - ACCURACY

Method	Mean	Standard Deviation
D. no adjustment	54.8059	5.3958
D. th=0.015, pr=1/170	77.6101	3.4103
D. w\ NLMS	14.9946	2.2053
D. th=0.01, pr=1/125	79.1496	2.9729126
E. with 0.5 s window	15.5043	0.6348
E. with 1s window	16.1568	0.6104
E. with 0.5 s 10 reps w\NLMS	74.7661	4.0303
E. with 0.5s , 10 reps	88.9543	2.0934
E. with 0.5s , 5 reps	88.0223	1.9547
E. with 0.5s , 5 reps w\NLMS	74.8302	4.7178
E. with 1s, 10 reps	90.4346	2.0785
E. with 1s , 10 reps w\NLMS	78.9243	4.9881
Hybrid method	92.5442	0.9285

V. CONCLUSIONS

The presented variations of the LMS algorithm have proven effective and accurate in modeling the complex EEG signal by overcoming temporary losses in stationarity and providing a reliable method of obtaining a precise time-frequency decomposition matrix. This gives way to continuing the work by expanding towards a complete BCI interface. Using the models thus obtained, the immediate development is to begin the detection of the subject's active band, ideally in real time. The powers for both rest and movement periods, evaluated in the

The presented variations of the LMS algorithm have been proven effective and accurate in modeling the complex EEG signal by overcoming temporary losses in stationarity and providing a reliable method of obtaining a precise time-frequency decomposition matrix. This gives wayto continuing the work by expanding towards a complete BCI interface. Thus, using the models obtained, the immediate development is to begin the detection of the subject's active band, ideally in real time. The powers for both rest and movement periods, evaluated in the said appropriate frequency interval will provide reliable feature vectors for classification algorithms. The accuracy of classification between left and right hand imagery will be the decisive factor in choosing which method is the most reliable for this task. The importance of on line processing in this situation must not be overlooked, but neither should be the possibility of developing a calibration protocol for the system that would insure stability for each subject, for further use.

We believe that improving the presented methods and working towards a complete BCI by expanding their functionality is a feasible objective that could bring forth advances in all related fields, including rehabilitation and prosthetic limb control.

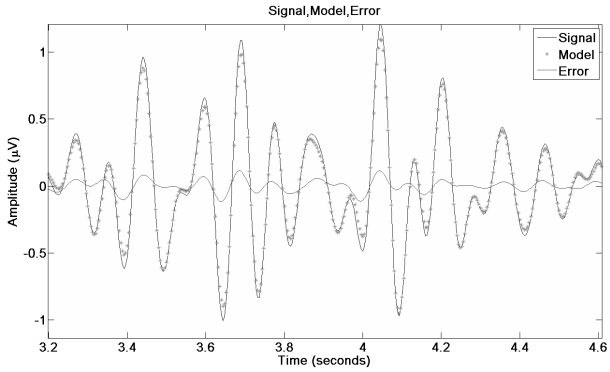

Fig. 2 The model obtained using method on a single left hand imagery trial for subject 3

VI. BIBLIOGRAPHY

[1] Y. Wang, K. Veluvoulu, J.-H. Cho and M. Defoort, "Adaptive estimation of EEG for subject-specific reactive band identification and improved ERD detection," *Neuroscience Letters,* no. 528, pp. 137-142, September 2012.

[2] C. Brunner, R. Leeb, G. R. M uller-Putz, A. Schl ogl and S. Pfurtscheller, " BCI Competition," Graz, 2008.

[3] K. Veluvolu, Y. Wang and S. S. Kavuri, "Adaptive estimation of EEG-rhythms for optimal band identification in BCI," *Journal of Neuroscience Methods,* no. 203, p. 163– 172, 2012.

[4] C. Vaz, X. Kong and N. Thakor, "An Adaptive Estimation of Periodic Signals Using a Fourier Linear COmbiner," *IEEE TRANSACTIONS ON SIGNAL PROCESSING,* vol. 42, no. 1, pp. 1-10, January 1994.

[5] G. Pfurtscheller and L. Da Silva, "Event-related EEG/MEG synchronization and desynchronization: basic principles," *Clinical Neurophysiology,* no. 110, pp. 1842-1857, May 1999.

[6] D. J. McFarland, L. M. McCane, S. V. David and J. R. Wolpaw, "Spatial filter selection for EEG-based communication," *Electroencephalography and clinical Neurophysiology,* no. 103, pp. 386-394, 1997.